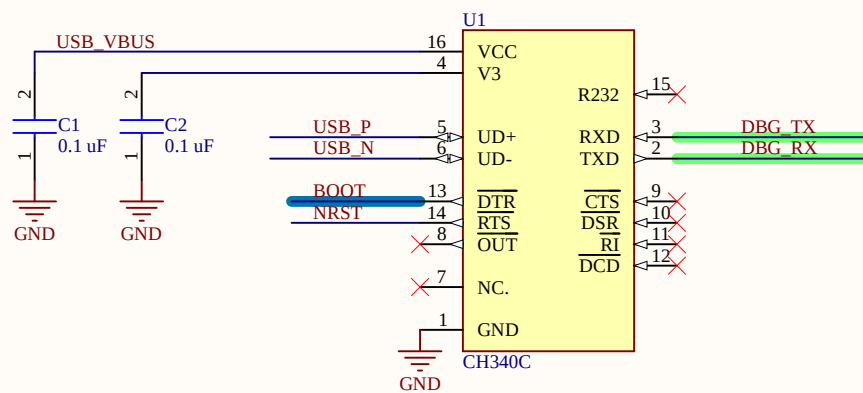
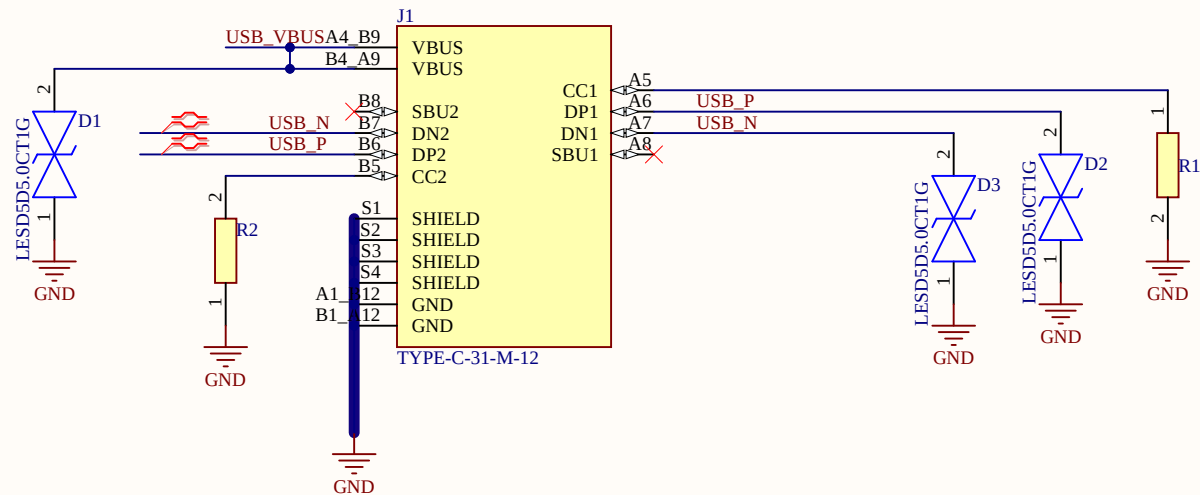




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[LCSC link](#)

Pin	IO	Function
1	NC	NC
2	NC	NC
3	3V3	3V3
4	EN	EN
5	SENSOR_VP	SENSOR_VP
6	SENSOR_VN	SENSOR_VN
7	IO34	IO34
8	IO35	IO35
9	IO32	IO32
10	IO33	IO33
11	IO25	IO25
12	IO26	IO26
13	IO27	IO27
14	IO12	IO12
15	IO13	IO13
16	NC	NC
17	NC	NC
18	NC	NC
19	NC	NC
20	NC	NC
21	NC	NC
22	NC	NC
23	IO15	IO15
24	IO2	IO2
25	ROOT	ROOT
26	SIM_EN	SIM_EN
27	SIM_RX	SIM_RX
28	SIM_TX	SIM_TX
29	BUZZER	BUZZER
30	DBG_TX	DBG_TX
31	DBG_RX	DBG_RX
32	I2C_1_SDA	I2C_1_SDA
33	I2C_1_CLK	I2C_1_CLK
34	GND	GND
35	GND	GND
36	GND	GND
37	GND	GND
38	GND	GND
39	GND	GND
40	GND	GND
41	GND	GND
42	GND	GND
43	GND	GND
44	GND	GND
45	GND	GND
46	GND	GND
47	GND	GND

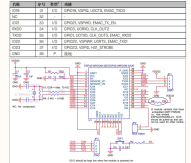
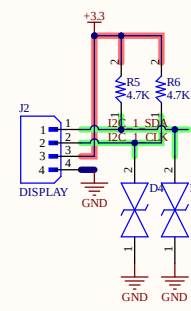
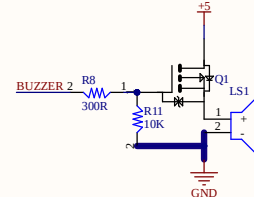
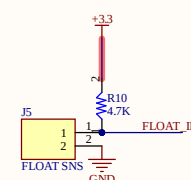
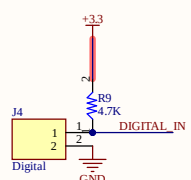
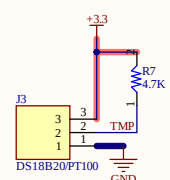
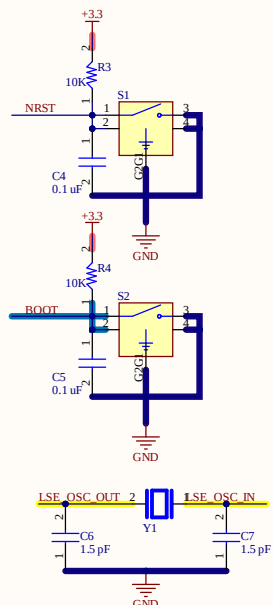
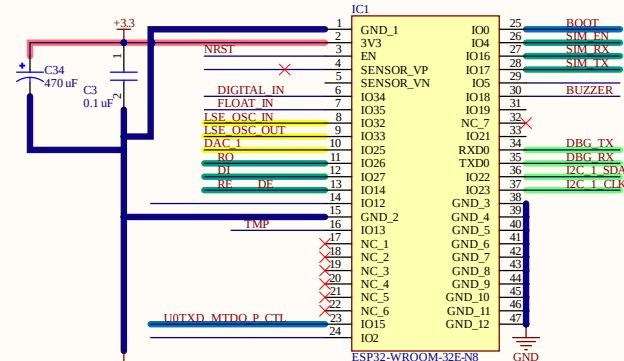


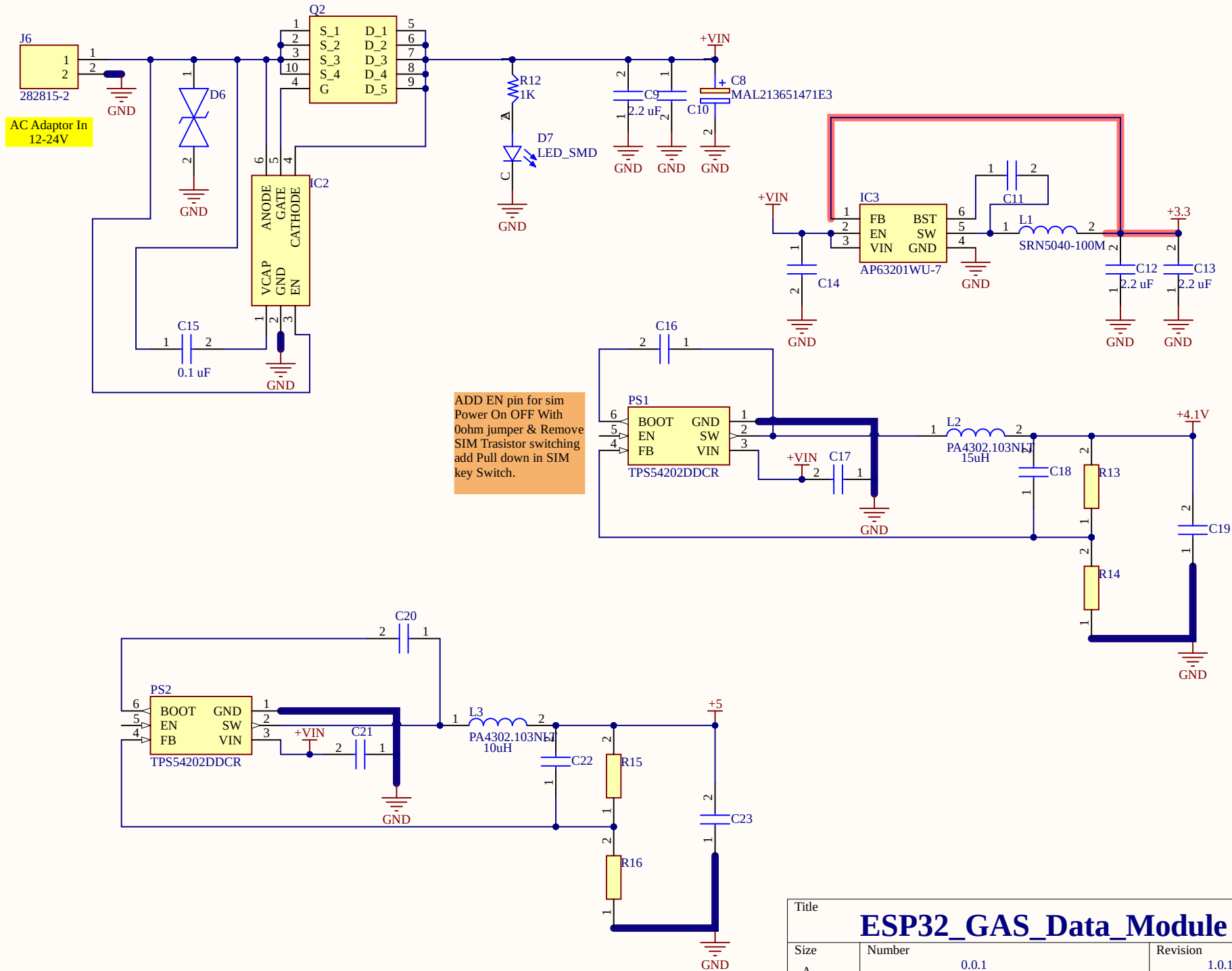
Table 4: Default Configuration of Strapping Pins

Strapping Pin	Default Configuration	Bit Value
GPIO0	Pull-up	1
GPIO2	Pull-down	0
MTDI	Pull-down	0
MTDO	Pull-up	1
GPIO5	Pull-up	1

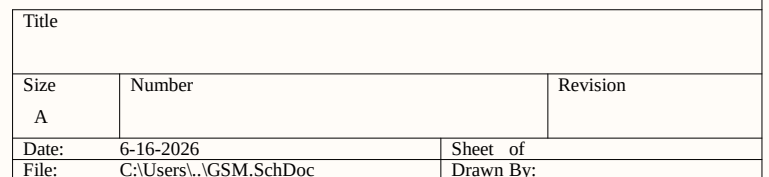
GPIO0	Pull-up	1
GPIO2	Pull-down	0
MTDI	Pull-down	0
MTDO	Pull-up	1
GPIO5	Pull-up	1

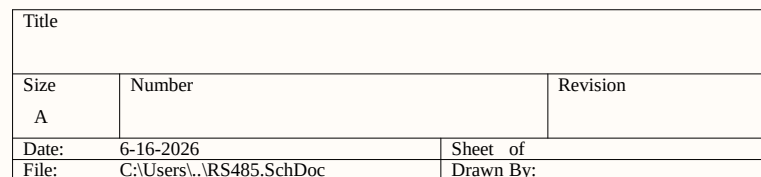


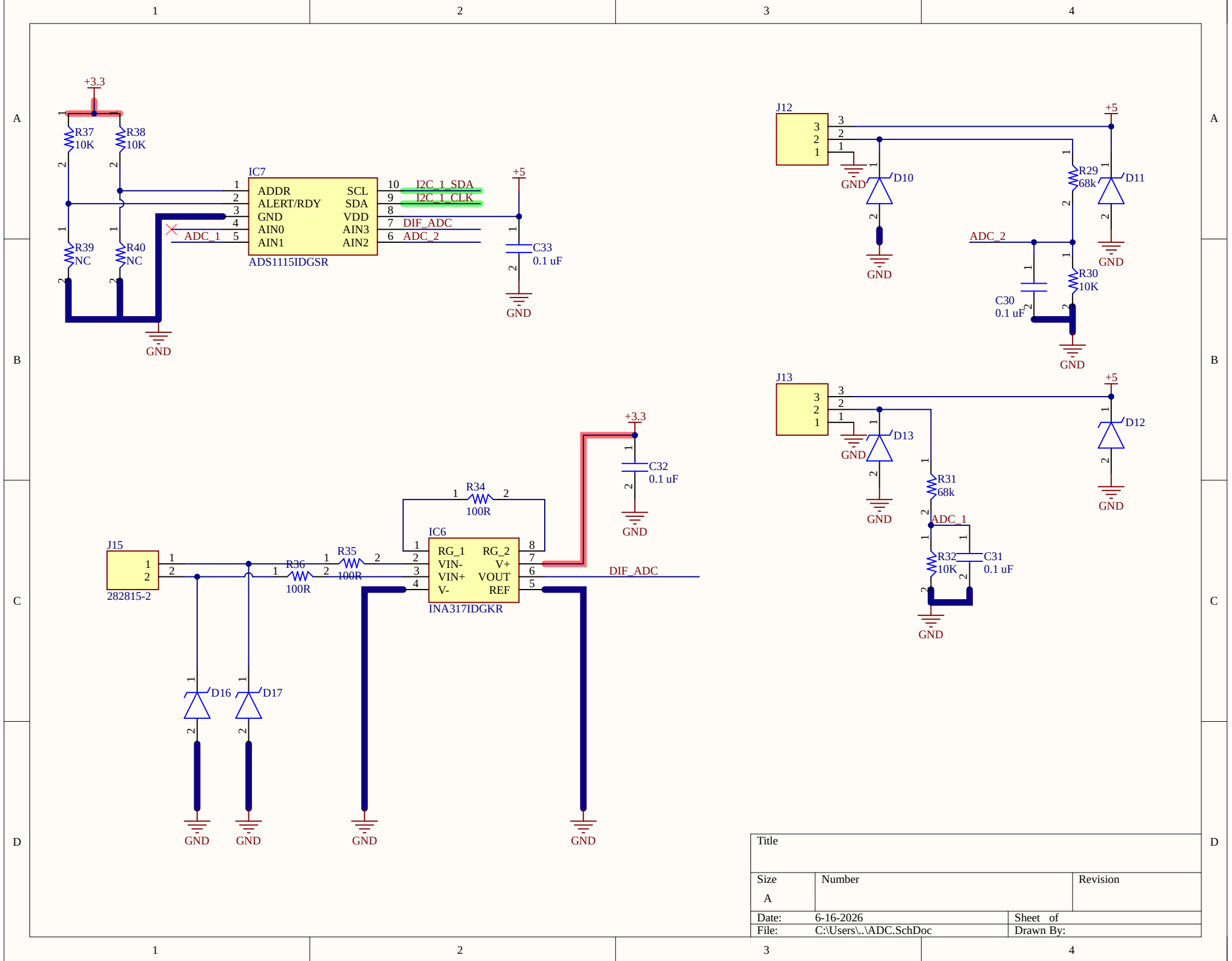
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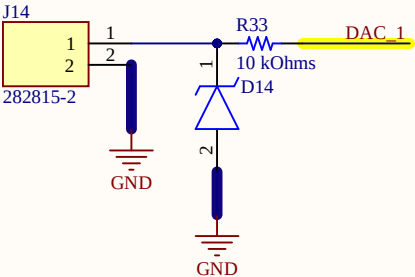
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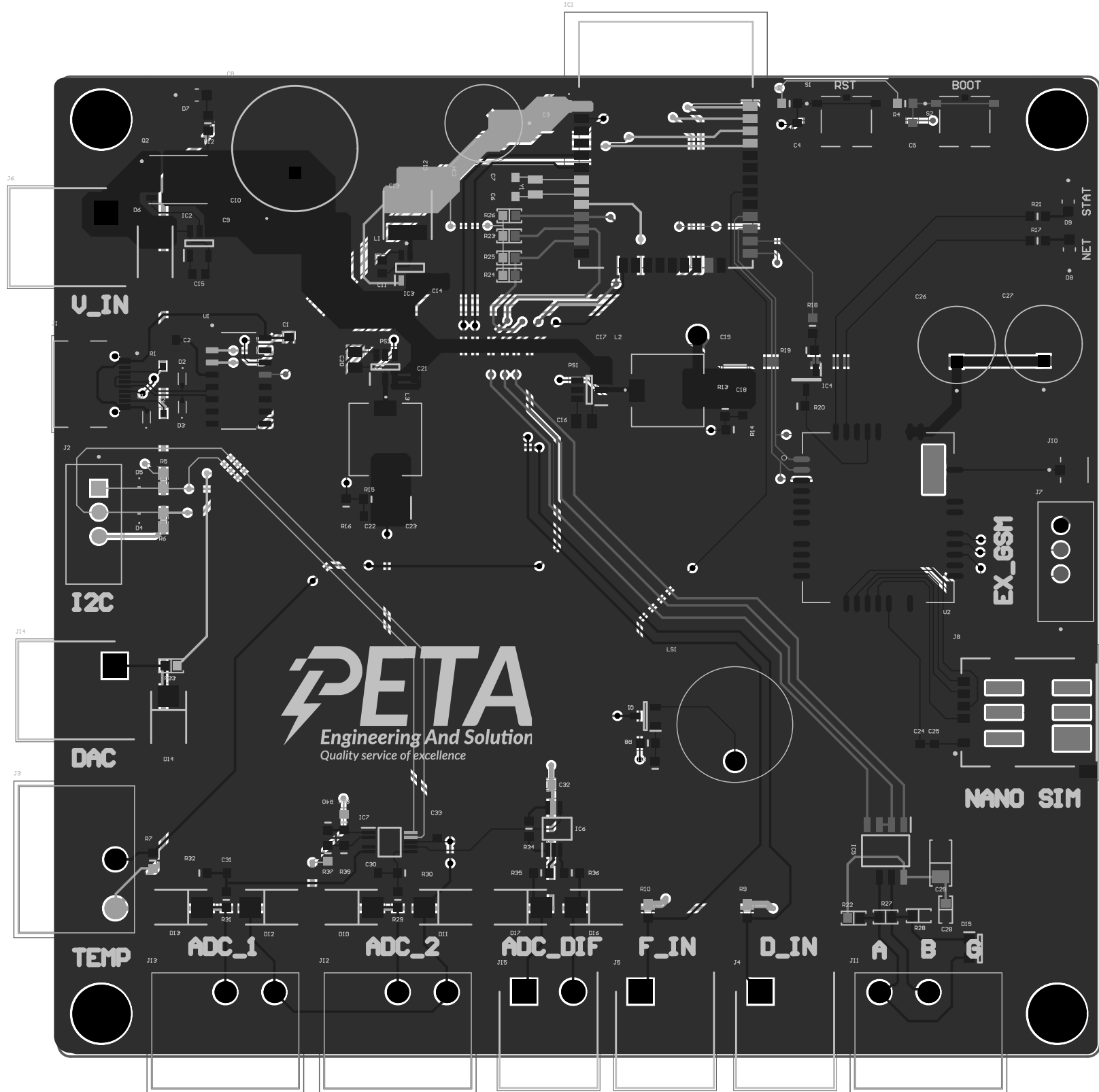




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Comment	Description	Designator	Footprint	LibRef	Quantity
0.1 uF	Capacitor	C1, C2, C3, C4, C5, C15, C20, C21, C32, C33	0408003	1000034707888104	10
1.5 uF	Capacitor	C8, C7	0408003	1000034707888104	2
ML213631471B3	Capacitor Polarised	C9	0408000M0001000H2700	ML213631471B3	1
2.2 uF	Capacitor	C9, C12, C13, C24, C25	0408003	1000034707888104	5
C5750SRH106K230K A	Capacitor	C10	C5750_Commercial	C5750SRH106K230K A	1
C385C104M69K7U	Capacitor	C11	0408003	C385C104M69K7U	1
C5750SRH106K230K A	Capacitor	C14, C17, C21	0408003	C5750SRH106K230K A	3
C385C104M69K7U	Capacitor	C16, C20	0408003	C385C104M69K7U	2
GM4035C1556J401 D	Capacitor	C18	0408003	GM4035C1556J401 D	1
GM4036R1G479K35 L	Capacitor	C19, C23	0408003	GM4036R1G479K35 L	2
C15A121CX11B01030 BA	Capacitor	C22	0408003	C15A121CX11B01030 BA	1
470 uF	220F 25V Aluminum Electrolytic Capacitors Radial, Can 2000 Hrs @ 105C	C26, C27, C34	040800350M500000 H1150	BLFC10221	3
0.1 uF	0408003 0.1UF50V X7R0003	C28	0408003_1608	04080030408003C7411	1
10 uF	0408003 10UF50V X5R 1206	C29	0408003_1206-1.8MM	12060030408003C7411	1
LESX06S.0CT1G	TVS Diode (Bi-directional)	D1, D2, D8	LESX06S.0CT1G	LESX06S.0CT1G	3
DESX03V321B3F-7	TVS Diode (Bi-directional)	D4, D5	DESX03V321B3F-7	DESX03V321B3F-7	2
9MBL24A-13-F	TVS Diode (Bi-directional)	D6	DIONM5306C250N	9MBL24A-13-F	1
LED_SMD	Red 635nm LED Indication - Discrete 2.1x0005 (2032 Metric)	D7, D8, D9	LEDX12500090-2N	9900130007F	3
9MBL5.0A	TVS Diode (Uni-directional)	D10, D11, D12, D13, D14, D16, D17	9MBL5.0A	9MBL5.0A	7
9M712	TVS Diode (Bi-directional)	D15	SDT06240X15-3N	9M712	1
ESP32-WROOM-32E-N8	Integrated Circuit	C1	ESP32-WROOM-32E-N8	ESP32-WROOM-32E-N8	1
LM74700QDB/TQ1	Integrated Circuit	IC2	SDT06280X145-6N	LM74700-Q1M8K-1_NCPB	1
AP63201MU-7	Integrated Circuit	IC3	SDT06280X140-6N	AP63201MU-7	1
MMBT2222ALT1G	Integrated Circuit	IC4	SDT06237X111-3N	MMBT2222ALT1G	1
MAX485ECS+	Integrated Circuit	IC5	SDT0627600X175-8N	MAX485ECS+	1
INA317IDGRR	Integrated Circuit	IC6	SDT06240X110-8N	INA317IDGRR	1
ADXL150DGR	Integrated Circuit	IC7	SDT06240X110-10N	ADXL150DGR	1
TYPEC-31-M-12	1.81 Connectors 38 Receptacle 1 8.94 7.3mm R4H4S	J1	HPD_TYPEC-31-M-12	TYPEC-31-M-12	1
DISPLAY	Connector Header Through Hole 4 position 0.098 (2.50mm)	J2	JST_B4B-XH-A(F)(SN)	B4B-XH-A(F)(SN)	1
DSB020 PT100	Connector	J3	2020153	2020153	1
Digital	Connector	J4	SDT06240X110-6N_P	202015-2	1
FLOAT_SNS	Connector	J5	SDT06240X110-6N_P	202015-2	1
202015-2	Connector	J6, J4, J5	SDT06240X110-6N_P	202015-2	3
DX_3M	Connector Header Through Hole 4 position 0.098 (2.50mm)	J7	JST_B4B-XH-A(F)(SN)	B4B-XH-A(F)(SN)	1
9MB000-6-1-14-00-A	6 Position Card Connector VARIO 3M Surface Mount, Right Angle Gold	J8	GCT_9MB000-6-1-14-00-A	9MB000-6-1-14-00-A	1
SMD_SMD	Connector Header Through Hole 4 position 0.098 (2.50mm)	J9	SMD_SMD	SMD_SMD	1
CONJL001-SMD	Connector	J10	CONJL001-SMD	CONJL001-SMD	1
202015-3	Connector	J11, J12, J13	2020153	202015-3	3
9N5040-100M	Inductor	L1	9N5040	9N5040-100M	1
PA4302-103NLT	Inductor	L2, L3	PA4302-103NLT	PA4302-103NLT	2
CM-1259FC-0585T	Loudspeaker or Buzzer	L3A	CM-1259FC-0585T	CM-1259FC-0585T	1
TPS54302DCOR	Power Supply	PS1, PS2	SDT06280X110-6N	TPS54302DCOR	2
CMC3415-1.7	Integrated Circuit	Q1	SDT06280X110-3N	CMC3415-1.7	1
RECU14N09NSATMAL	DC/DC (N-Channel)	Q2	RECU14N09NSATMAL	RECU14N09NSATMAL	1
CHV0402100KFR04-P	Resistor	R1, R2	RES1005X04N	CHV0402100KFR04-P	2
10K	Resistor	R3, R4, R11, R20, R22, R27, R28	RES0603	RES	7
4.7K	Resistor	R5, R6, R7, R8, R10, R18	RES0603	RES	6
100K	Resistor	R9	RES0603	RES	1
1K	Resistor	R12, R17, R21	RES0603	RES	3
CHV0402100KFR04-P	Resistor	R13, R15	RES0603	CHV0402100KFR04-P	2
CHV0402100KFR04-P	Resistor	R14	RES0603	CHV0402100KFR04-P	1
CHV0402100KFR04-P	Resistor	R16	RES0603	CHV0402100KFR04-P	1
1K	Resistor	R19	RES0603	RES	1
10K	Resistor	R20, R26, R40	RES0603	RES	3
20 kOhms	RES SMD 10K OHM 1% 1/10W 0603	R22, R28	RES0603_1608	RT0603PF0710K	2
10 kOhms	RES SMD 10K OHM 1% 1/10W 0603	R23, R24, R25, R26, R27	RES0603_1608	RT0603PF0710K	5
120	RES SMD 10K OHM 1% 1/10W 0603	R27	RES0603_1608	RT0603PF0710K	1
56K	Resistor	R29, R31	RES0603	RES	2
100K	Resistor	R34, R35, R36	RES0603	RES	3
PT3206_5MG15_SMT8_Z_LFS	Toggle Switch SPST-NO Tact Actuated Surface Mount	SL, S2	SW_PT3206_5MG15_SMT8_Z_LFS	PT3206_5MG15_SMT8_Z_LFS	2
0-G40C	USB to serial chip C1540	U1	SDT0627600X180-10N	0-G40C	1
9MB00C	Module GSM/Bluetooth; GPS; 2G, 1800MHz; 300MHz; 85.9kbaud; SMD	U2	LC1110P1570X1780X90-42N	9MB00C	1
CMV-T1A-32-76894-6-PPF-20PM-TA-QC	Tuning Fork Crystal 32.768kHz in SMD L2 x 2.0mm package	V1	XTAL_CMV-T1A-32.76894-6-PPF-20PM-TA-QC	CMV-T1A-32.76894-6-PPF-20PM-TA-QC	1